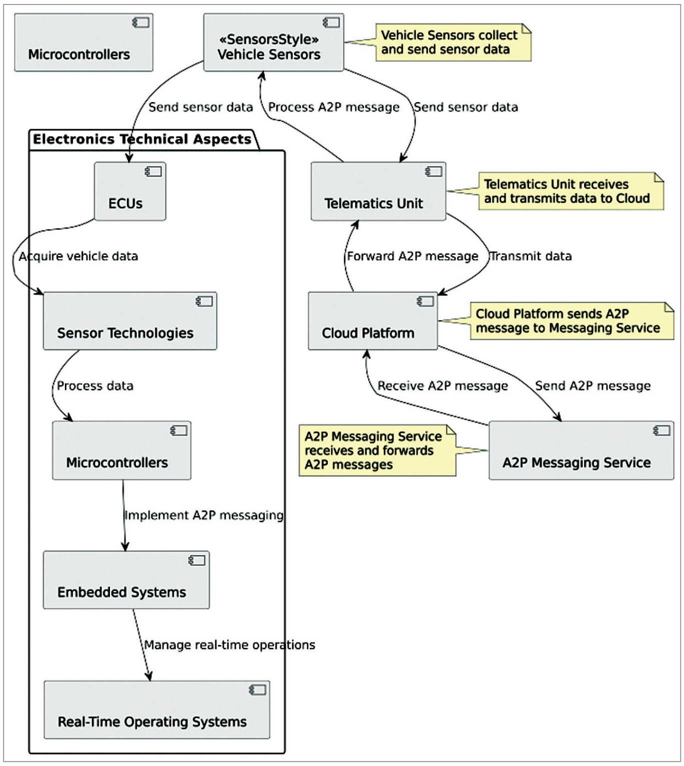
• *Engine start/stop.* Drivers can use A2P messaging to remotely start or stop their vehicles, offering convenience and climate control

3. Over-the-air (OTA) updates

• *Software updates.* A2P messaging facilitates the seamless delivery of OTA updates for in-vehicle software, ensuring that vehicles stay up-to-date with the latest features and security patches

• *Firmware upgrades.* Connected vehicles leverage A2P messaging to update firmware, improving the performance and capabilities of various onboard systems

4. In-vehicle entertainment and information



Working of A2P messaging in connected vehicles

• *Personalised recommendations.* A2P messaging delivers tailored recommendations for in-vehicle entertainment options, such as music, podcasts, or nearby points of interest

• *Traffic and navigation updates.* Drivers receive real-time traffic updates and navigation suggestions, optimising routes for a smoother driving experience

Benefits of A2P messaging

A2P messaging plays a crucial role in the context of connected vehicles, providing various benefits that improve communication, safety, and user experience. Here are some key advantages:

1. Instant alerts and notifications. Receive real-time updates on traffic, weather, and maintenance, empowering drivers to make informed decisions on the go

2. Fleet management made easy. A2P messaging streamlines operations for businesses with vehicle fleets, optimising routes and scheduling maintenance seamlessly

3. Proactive driver assistance. Enjoy personalised suggestions for fuel-efficient routes, nearby amenities, and parking availability, increasing the overall driving experience

4. Emergency assistance and safety alerts. Automatic distress signals in emergencies ensure swift response, potentially saving lives and minimising road incidents

5. Convenient vehicle services. Manage vehicle well-being effortlessly, from scheduling maintenance appointments to receiving real-time updates on repairs

6. Minimised distractions. A2P messaging delivers crucial information without distracting drivers, keeping their focus on the road

7. Data-driven continuous improvement. Harness valuable data for analytics, refining offerings, improving safety features, and tailoring services to drivers' evolving needs

Main components of A2P messaging

Messaging in connected vehicles involves communication between applications and users within the context of vehicle connectivity. This type of messaging is becoming increasingly important for various applications in connected vehicles, including safety alerts, navigation assistance, and vehicle diagnostics. Here are the main components:

1. Telematics unit. The telematics unit is a crucial component of connected vehicles. It comprises hardware and software that enable communication between the vehicle and external entities. This unit often includes sensors, GPS modules, and communication interfaces.

2. Connectivity. A2P messaging relies on robust connectivity, such as cellular networks (3G, 4G, and increasingly 5G) or wireless technologies like Wi-Fi. This ensures that the vehicle can communicate with external servers and applications in real time.

3. Messaging gateway. The messaging gateway acts as an intermediary between the vehicle and external applications. It manages the flow of messages, ensuring that they are delivered securely and efficiently. It may include protocols for message encoding and decoding.

4. Cloud platform. A cloud-based platform is often used to store and process the data exchanged between the vehicle and external applications. This platform facilitates the storage of vehicle data, analytics, and the deployment of applications or services.

5. Application servers. Application servers host the specific A2P applications that communicate with